

5.3 ASEXUAL REPRODUCTION

When an offspring is produced by a single parent without the involvement of gamete formation, the reproduction is **asexual**. In this method, a single individual (parent) is capable of producing the offspring. As a result, the offspring are not only identical to one another but also exact copies of their parent. The term '**clone**' is used to describe such morphologically and genetically similar individuals. Asexual reproduction is widespread among different organisms. It is common in members of Protista, Bacteria, Archaea and in multicellular organisms with relatively simple organisation. The offspring show '**uniparental inheritance**' without any genetic variations.

5.3.1 Asexual Reproduction in Protozoans

The methods of asexual reproduction that occur in many protozoans are binary fission and multiple fission.

5.3.2 Binary Fission

It is the most common method of asexual reproduction in protozoans. It occurs during **favourable conditions**. It is the division of the parent cell into two daughter individuals. It involves the division of nucleus (karyokinesis) followed by the division of the cytoplasm (cytokinesis). The plane of fission differs in different protozoans. Based on the axis of cytokinesis, the binary fission is chiefly classified into longitudinal and transverse fissions.

I. Longitudinal Binary Fission

It is seen in flagellates such as ***Euglena***, ***Trypanosoma*** etc. In this type of binary fission, the body divides into two halves longitudinally, hence called longitudinal **binary fission**.

Euglena possesses two flagella arising from two basal granules, one contractile vacuole, stigma, paraflagellar body (photoreceptor), cytostome, cytopharynx and reservoir anteriorly, nucleus in the centre and chromatophores around the nucleus and myonemes beneath the proteinous pellicle.

During the process of binary fission, the nucleus, basal granules, chromatophores, cytoplasm undergo division. The nucleus divides by **mitosis** into two daughter nuclei. Then the kinetosomes and the chromatophores also divide. At first, a longitudinal groove develops in the middle of the anterior end. This groove extends gradually towards the posterior end until

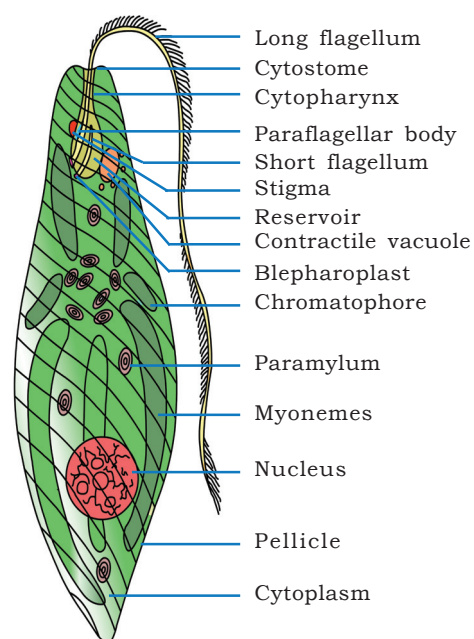


Fig. 5.9 *Euglena*

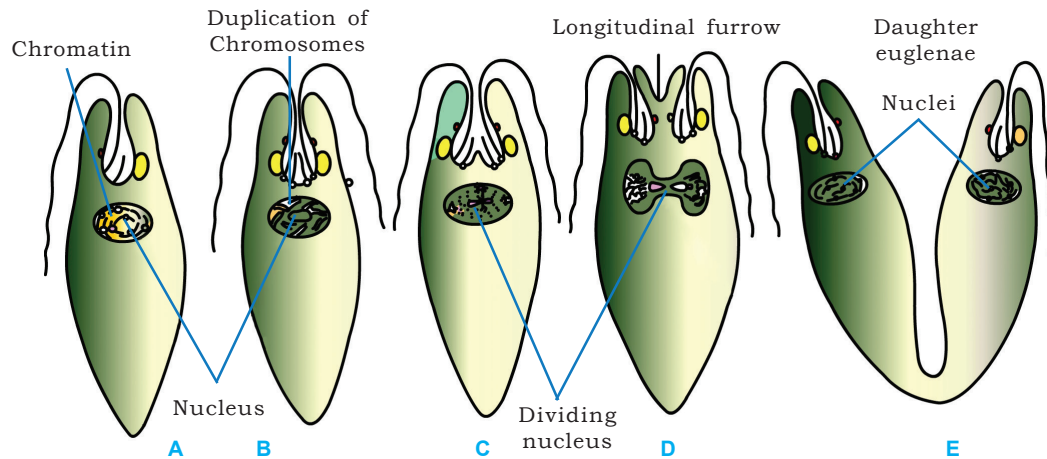


Fig. 5.7 Longitudinal binary fission in *Euglena* - different stages

the two daughter individuals are separated. One daughter *Euglena* retains the parental flagella. The other daughter individual develops new flagella from the newly formed basal granules. The stigma, paraflagellar body and contractile vacuole of the parent disappear. They develop afresh in both the daughter euglenae. **The longitudinal binary fission is known as symmetrogenic division**, because the two daughter euglenae resemble each other like ‘**mirror images**’.

II. Transverse Binary Fission

It is performed by *Paramecium*, commonly called “slipper animalcule”. It has an oral groove, cytostome and cytopharynx in the oral surface. It also has one macronucleus (**polyploid**) and one micronucleus (**diploid**), two contractile vacuoles (anterior and posterior), trichocysts, infraciliary system and many cilia on the body.

During favourable conditions, *Paramecium* stops feeding after attaining its

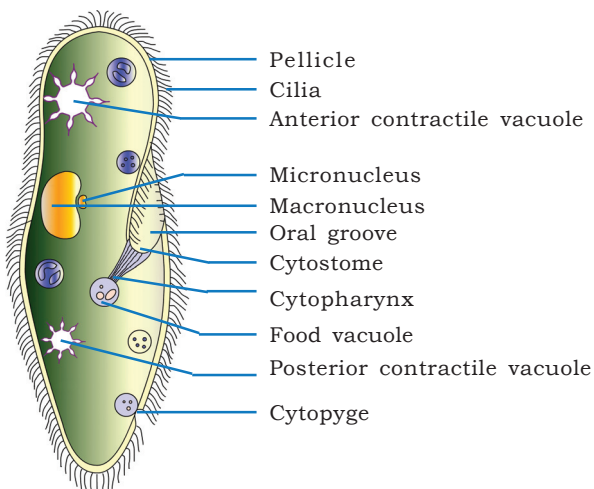


Fig 5.8 *Paramecium*

maximum growth. At first the micronucleus divides by **mitosis** and the macronucleus divides into two daughter nuclei by **amitosis**. The oral groove disappears. After karyokinesis, a transverse constriction appears in the middle of the body, which deepens and divides the parent cell into two daughter individuals, the anterior **proter** and the posterior **opisthe**. The proter receives the anterior contractile vacuole, cytopharynx and cytosome from its parent individual. It develops posterior contractile vacuole and a new oral groove. The **opisthe** receives the posterior contractile vacuole of its parent. It

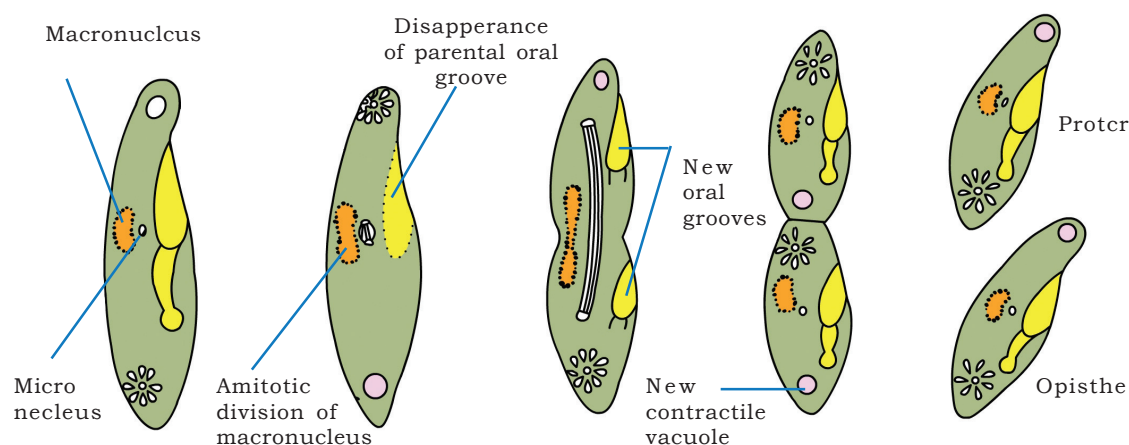


Fig. 5.9 Stages of transverse binary fission in *Paramecium*

develops a new anterior contractile vacuole, cytopharynx, cytostome and a new oral groove. Binary fission is completed in almost two hours, in favourable conditions and ***Paramecium*** can produce four generations of daughter individuals by binary fission in a day.

The transverse binary fission is also called **homothetogenic fission**, because the plane of fission is at right angles to the longitudinal axis of the body. As it occurs at right angles to the kineties, it is also called **perkinetal** fission (across the Kinetia).

5.3.3 Multiple Fission

It is the division of the parent body into many smaller individuals (Multi – many; Fission – splitting). Normally multiple fission occurs during **unfavourable** conditions. During the multiple fission, the nucleus first undergoes repeated mitotic divisions without cytokinesis. This causes the formation of many daughter nuclei. Then the cytoplasm also divides into as many number of bits as there are nuclei. Each cytoplasmic bit encircles one daughter nucleus. This results in the formation of many smaller individuals from a single parent organism.

There are different types of multiple fissions in protozoans such as **schizogony**, **male gametogony**, **sporogony** in ***Plasmodium***, **sporulation** in ***Amoeba***, etc.

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